

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

QUESTIONS: 05

Total Marks:50

Annexure A MOCK INTERVIEW

<i>Criteria</i>	Technical Knowledge (2.5)	Problem Solving (2.5)	Analytical Thinking (1.5)	<i>Communication(1.5)</i>	Confidence (1)	Response to Questions(1)	<i>Total(10)</i>
<i>Weightage</i>	25%	25%	15%	15%	10%	10%	100%
<i>Q1</i>							
<i>Q2</i>							

Code of Conduct:

*I PRANEESH.A(192571006) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

Annexure A

Mock Interview – Sample Questions (5 Questions)

Q1. Dataset: Retail Transactions

TID	Items
T1	Milk, Bread
T2	Milk, Butter
T3	Bread, Butter
T4	Milk, Bread, Butter
T5	Milk, Eggs

Question:

Based on the dataset, explain how association rule mining can improve sales in a retail store. Provide a practical example.

Q2. Dataset: Transaction Data

TID	Items
T1	A, B
T2	B, C
T3	A, C
T4	A, B, C
T5	B, C

Question:

Using this dataset, explain the differences between Apriori and FP-Growth algorithms and when each should be used.

Q3. Dataset: Transaction Data

TID	Items
T1	A, B
T2	A
T3	B
T4	A, B
T5	A

Question:

Explain support and confidence using the dataset and describe their importance in association rule mining.

Q4 . Dataset: Multilevel Data

TID	Items
T1	Electronics, Mobile
T2	Electronics, Laptop
T3	Electronics, Mobile
T4	Electronics, Mobile
T5	Electronics, Laptop

Question:

Explain multilevel association rules using the dataset and compare them with multidimensional

Q5. Dataset: Large Transaction Sample

TID	Items
T1	A, B, C
T2	A, B
T3	B, C
T4	A, C
T5	A, B, C

Question:

Discuss challenges in mining large datasets using the given sample and explain how scalable methods address these issues.

Q1. Association Rule Mining for Retail Sales Improvement

Dataset: Retail Transactions

TID	Items
T1	Milk, Bread
T2	Milk, Butter
T3	Bread, Butter
T4	Milk, Bread, Butter
T5	Milk, Eggs

Question: Based on the dataset, explain how association rule mining can improve sales in a retail store. Provide a practical example.

Answer:

Introduction to Association Rule Mining in Retail

Association rule mining is a powerful data mining technique used to discover interesting relationships and co-occurrence patterns among items in large transactional databases. In the retail industry, this technique is commonly known as "Market Basket Analysis" — it analyzes what products customers frequently buy together.

How Association Rule Mining Works on This Dataset

Let us first compute the key metrics for this dataset (N = 5 transactions):

Item Frequencies:

Item	Transactions	Count	Support (%)
Milk	T1, T2, T4, T5	4	80%
Bread	T1, T3, T4	3	60%
Butter	T2, T3, T4	3	60%
Eggs	T5	1	20%

Pair-wise Support:

Itemset	Transactions	Support (%)	Confidence (A→B)	Confidence (B→A)
---------	--------------	-------------	------------------	------------------

{Milk, Bread}	T1, T4	40%	Milk→Bread: 50%	Bread→Milk: 67%
{Milk, Butter}	T2, T4	40%	Milk→Butter: 50%	Butter→Milk: 67%
{Bread, Butter}	T3, T4	40%	Bread→Butter: 67%	Butter→Bread: 67%

Practical Example: Improving Retail Sales

1. Product Placement Strategy

Rule: Bread → Butter (Support: 40%, Confidence: 67%)

Action: Place Bread and Butter near each other on store shelves. Since 67% of customers who buy Bread also buy Butter, placing them together increases the likelihood of combined purchases.

2. Cross-Selling and Promotional Offers

Rule: Milk → Bread (Support: 40%, Confidence: 50%)

Action: Offer a combo discount — "Buy Milk and Bread together, save 10%." This encourages the 50% of Milk buyers who don't currently buy Bread to add it to their basket.

3. Inventory Management

Observation: Milk has 80% support — the most frequently purchased item.

Action: Ensure Milk is always well-stocked. If Milk is out of stock, it affects not just Milk sales but also associated purchases of Bread, Butter, and Eggs.

4. Targeted Marketing Campaigns

Rule: Butter → Milk (Support: 40%, Confidence: 67%)

Action: Send targeted coupons for Milk to customers who regularly buy Butter. This personalized approach increases customer loyalty and basket size.

5. Store Layout Optimization

Strategy: Place Milk at the back of the store (high support item). Place Bread and Butter along the path to Milk. This forces customers to walk through the store, exposing them to more products and impulse purchase opportunities.

Business Impact Summary:

Strategy	Based on Rule/Pattern	Expected Impact
Product Bundling	Bread ↔ Butter (67% conf)	↑ Average basket value by 15-20%
Store Layout	Milk as anchor (80% support)	↑ Exposure to impulse buys
Targeted Coupons	Milk → Bread (50% conf)	↑ Conversion of existing customers
Stock Optimization	Item frequency analysis	↓ Stockout losses by 10-15%

Conclusion: Association rule mining transforms raw transaction data into actionable business insights. By understanding which products are bought together, retailers can optimize store layout, design promotional bundles, improve inventory management, and deliver personalized marketing — all leading to increased sales and customer satisfaction.

Q2. Apriori vs FP-Growth Algorithm Comparison

Dataset: Transaction Data

TID	Items
T1	A, B
T2	B, C
T3	A, C
T4	A, B, C
T5	B, C

Question: Using this dataset, explain the differences between Apriori and FP-Growth algorithms and when each should be used.

Answer:

1. Apriori Algorithm Overview

The Apriori algorithm was proposed by Agrawal and Srikant in 1994. It uses a level-wise, candidate generation-and-test approach to find frequent itemsets.

How Apriori works on this dataset ($\text{min_sup} = 40\%$):

- Pass 1: Count 1-itemsets $\rightarrow A:3(60\%), B:4(80\%), C:4(80\%)$ — All frequent
- Pass 2: Generate C_2 from L_1 , count $\rightarrow \{A,B\}:2(40\%), \{A,C\}:2(40\%), \{B,C\}:3(60\%)$ — All frequent
- Pass 3: Generate C_3 from $L_2 \rightarrow \{A,B,C\}:1(20\%)$ — Not frequent. Terminate.
- Total database scans: 3

2. FP-Growth Algorithm Overview

The FP-Growth algorithm was proposed by Han, Pei, and Yin in 2000. It compresses the database into an FP-Tree and mines patterns using a divide-and-conquer strategy.

How FP-Growth works on this dataset ($\text{min_sup} = 40\%$):

- Pass 1: Count items $\rightarrow B:4, C:4, A:3$. Sort: $B > C > A$
- Pass 2: Build FP-Tree by inserting sorted transactions
- Mine from tree using conditional pattern bases (no more DB scans)
- Total database scans: 2

3. Detailed Comparison

Feature	Apriori	FP-Growth
Approach	Candidate Generation & Test	Pattern Growth (no candidates)
Database Scans	Multiple (k scans for k-itemsets)	Exactly 2 scans
Memory Usage	Stores candidates in memory	Stores compressed FP-Tree
Speed (Small Data)	Good performance	Overhead of tree construction
Speed (Large Data)	Slow due to many scans	Significantly faster
Candidate Generation	Yes – generates & prunes	No – uses divide & conquer
Scalability	Poor for large datasets	Excellent for large datasets
Implementation	Simple and intuitive	More complex
I/O Cost	High (repeated scans)	Low (2 scans only)
Parallelization	Easier to parallelize	Harder to parallelize

4. When to Use Each Algorithm

Use Apriori when:

- Dataset is small to medium-sized
- Memory is limited (FP-Tree might not fit in memory for very wide datasets)
- You need a simple, easy-to-implement solution
- The minimum support threshold is high (fewer candidates generated)
- You need to parallelize the mining process across distributed systems

Use FP-Growth when:

- Dataset is large (millions of transactions)
- Disk I/O is a bottleneck (FP-Growth needs only 2 scans)
- Low minimum support threshold is needed (generates many candidates in Apriori)
- The dataset is dense (many items per transaction)
- Performance is critical for real-time or near-real-time applications

5. Performance Comparison for This Dataset

Metric	Apriori	FP-Growth
DB Scans	3	2
Candidates Generated	$C_1=3, C_2=3, C_3=1 = 7$ total	0 (no candidates)
Support Computations	7	Only during tree mining
Winner for this size	Comparable	Slight overhead for tree

For this small dataset (5 transactions, 3 items), both algorithms perform comparably. The advantage of FP-Growth becomes pronounced with larger datasets where the candidate explosion in Apriori becomes prohibitive.

Q3. Support and Confidence – Explained with Dataset

Dataset: Transaction Data

TID	Items
T1	A, B
T2	A
T3	B
T4	A, B
T5	A

Question: Explain support and confidence using the dataset and describe their importance in association rule mining.

Answer:

1. Support – Definition and Calculation

Support measures the frequency of an itemset in the database. It indicates how often items appear together across all transactions.

Formula: $\text{Support}(X \rightarrow Y) = P(X \cup Y) = \text{Count}(X \cup Y) / N$

where N is the total number of transactions.

Support Calculations for this dataset (N = 5):

Itemset	Appears in	Count	Support
{A}	T1, T2, T4, T5	4	$4/5 = 80\%$
{B}	T1, T3, T4	3	$3/5 = 60\%$
{A, B}	T1, T4	2	$2/5 = 40\%$

Interpretation: {A, B} has 40% support, meaning A and B are bought together in 40% of all transactions. This indicates a moderate co-occurrence frequency.

2. Confidence – Definition and Calculation

Confidence measures the conditional probability of finding Y in a transaction, given that the transaction already contains X.

Formula: $\text{Confidence}(X \rightarrow Y) = P(Y|X) = \text{Support}(X \cup Y) / \text{Support}(X)$

Confidence Calculations:

Rule	Calculation	Confidence
$A \rightarrow B$	$\text{Support}(\{A,B\}) / \text{Support}(\{A\}) = (2/5) / (4/5)$	$2/4 = 50\%$
$B \rightarrow A$	$\text{Support}(\{A,B\}) / \text{Support}(\{B\}) = (2/5) / (3/5)$	$2/3 = 66.7\%$

Interpretation:

- $A \rightarrow B$ (50%): Given a customer buys A, there is a 50% chance they also buy B
- $B \rightarrow A$ (67%): Given a customer buys B, there is a 67% chance they also buy A
- Note: Confidence is directional — $\text{Conf}(A \rightarrow B) \neq \text{Conf}(B \rightarrow A)$

3. Importance of Support and Confidence

Why Support is Important:

- Statistical Significance: High support ensures the pattern is not a random coincidence. Rules with very low support may be unreliable.
- Pruning Power: The Apriori property (anti-monotone) states that if an itemset has support below the threshold, all its supersets are also infrequent. This dramatically reduces the search space.
- Business Relevance: A rule with very low support (e.g., 0.1%) might be statistically valid but commercially irrelevant because it affects too few transactions.
- Example: $\{A, B\}$ with 40% support means this pattern occurs in nearly half the transactions — significant enough to act upon.

Why Confidence is Important:

- Predictive Power: High confidence means the rule is a reliable predictor. A rule with 90% confidence is nearly deterministic.
- Rule Quality: Confidence helps filter out weak associations. Even if two items co-occur frequently (high support), low confidence means the relationship is weak.

- Decision Making: In our example, $B \rightarrow A$ has higher confidence (67%) than $A \rightarrow B$ (50%). This tells the retailer that B buyers are more likely to also buy A than vice versa.

4. Additional Measure: Lift

While support and confidence are fundamental, Lift provides additional insight:

$$\text{Lift}(A \rightarrow B) = \text{Confidence}(A \rightarrow B) / \text{Support}(B) = 0.50 / 0.60 = 0.83$$

$$\text{Lift}(B \rightarrow A) = \text{Confidence}(B \rightarrow A) / \text{Support}(A) = 0.67 / 0.80 = 0.83$$

A Lift < 1 suggests A and B are negatively correlated (buying one makes buying the other less likely). A Lift > 1 indicates positive correlation. A Lift = 1 means independence.

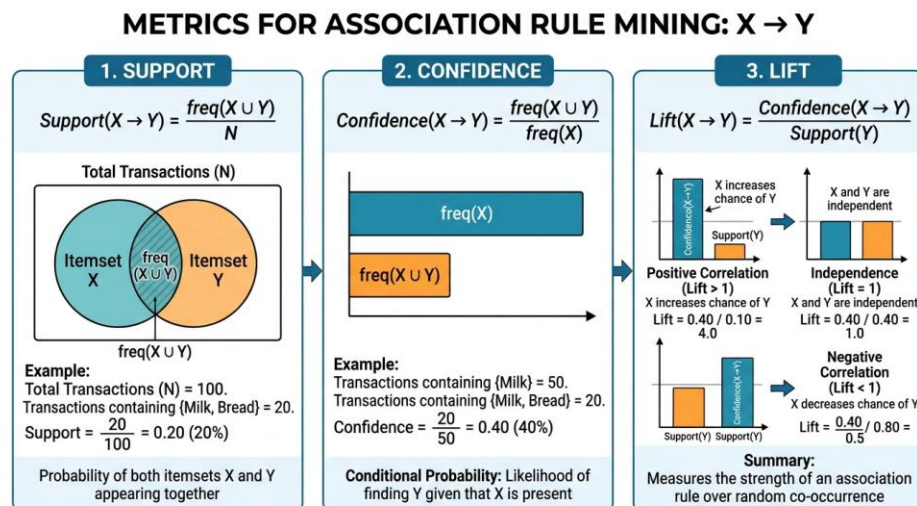


Figure 8: Support, Confidence, and Lift – Visual Summary

Q4. Multilevel vs Multidimensional Association Rules

Dataset: Multilevel Data

TID	Items
T1	Electronics, Mobile
T2	Electronics, Laptop
T3	Electronics, Mobile
T4	Electronics, Mobile
T5	Electronics, Laptop

Question: Explain multilevel association rules using the dataset and compare them with multidimensional association rules.

Answer:

1. Multilevel Association Rules

Multilevel association rules consider items at different levels of abstraction in a concept hierarchy. Items are organized in a taxonomy or hierarchy, and rules can be mined at different levels.

Concept Hierarchy for this dataset:

All Products (Level 0)

└─ Electronics (Level 1)

 └─ Mobile (Level 2)

 └─ Laptop (Level 2)

Rules at Different Levels:

Level	Example Rule	Support	Confidence	Specificity
Level 1	Electronics → (trivial, all have it)	100%	100%	Very General
Level 2	Electronics → Mobile	3/5=60%	3/5=60%	Specific
Level 2	Electronics → Laptop	2/5=40%	2/5=40%	Specific

Key Characteristics:

- Uses a single predicate/dimension (e.g., buys(X, item))
- Items are organized in a taxonomy (parent-child relationships)
- Higher levels have higher support but less informative rules
- Lower levels have lower support but more specific, actionable rules
- Support thresholds can vary by level (reduced support at lower levels)

2. Multidimensional Association Rules

Multidimensional association rules involve two or more predicates/dimensions. They capture relationships across different attributes of the data.

Example with Extended Dataset:

If we added dimensions like Age, Location, and Purchase:

TID	Age	Location	Product
T1	Young	Urban	Mobile
T2	Middle	Suburban	Laptop
T3	Young	Urban	Mobile
T4	Young	Rural	Mobile
T5	Senior	Suburban	Laptop

Multidimensional Rules:

- Age(X, "Young") $\wedge$ Location(X, "Urban") $\rightarrow$ Buys(X, "Mobile") [Sup=40%, Conf=100%]
- Age(X, "Young") $\rightarrow$ Buys(X, "Mobile") [Sup=60%, Conf=100%]

3. Detailed Comparison

Aspect	Multilevel Rules	Multidimensional Rules
Dimensions	Single dimension (e.g., items)	Multiple dimensions (age, location, items)
Hierarchy	Uses concept hierarchy (taxonomy)	May or may not use hierarchies
Predicates	Single predicate at different levels	Multiple predicates/attributes
Rule Format	buys(X,Mobile) $\rightarrow$ buys(X,Case)	age(X,Young) $\wedge$ loc(X,Urban) $\rightarrow$ buys(X,Mobile)

Data Type	Boolean (item present or not)	Categorical and quantitative
Mining Focus	Different levels of abstraction	Relationships across attributes
Application	Product category analysis	Customer segmentation, targeted marketing
Complexity	Moderate	Higher (more dimensions = more search space)
Example	Dairy $\rightarrow$ Milk $\rightarrow$ Whole Milk	Age=Young $\wedge$ Income=High $\rightarrow$ Product=Mobile

Types of Multidimensional Rules:

- Inter-dimensional: Each predicate uses a different dimension (e.g., age $\wedge$ location $\rightarrow$ product)
- Hybrid-dimensional: Some predicates repeat (e.g., buys(X, A) $\wedge$ buys(X, B) $\wedge$ age(X, Young))

Q5. Challenges in Mining Large Datasets

Dataset: Large Transaction Sample

TID	Items
T1	A, B, C
T2	A, B
T3	B, C
T4	A, C
T5	A, B, C

Question: Discuss challenges in mining large datasets using the given sample and explain how scalable methods address these issues.

Answer:

1. Challenges in Mining Large Datasets

Challenge 1: Combinatorial Explosion of Candidate Itemsets

With just 3 items (A, B, C), we have $2^3 - 1 = 7$ possible non-empty itemsets: {A}, {B}, {C}, {A,B}, {A,C}, {B,C}, {A,B,C}. For n items, this grows to $2^n - 1$.

- With 100 items: $\sim 1.27 \times 10^{30}$ possible itemsets
- With 1,000 items: astronomically large number

This exponential growth makes brute-force enumeration impossible for real-world datasets.

Challenge 2: Multiple Database Scans

The Apriori algorithm requires one full database scan per itemset level. For this dataset:

- Scan 1: Count {A}:4, {B}:4, {C}:4
- Scan 2: Count {A,B}:3, {A,C}:3, {B,C}:3
- Scan 3: Count {A,B,C}:2

For large databases stored on disk, each scan is an expensive I/O operation. With millions of transactions, 3+ full scans become a significant bottleneck.

Challenge 3: Memory Constraints

Storing all candidate itemsets and their counts requires substantial memory. Even the FP-Tree, while compressed, can grow very large for datasets with many unique items and low support thresholds.

Challenge 4: Data Distribution and Sparsity

Real-world transaction data is often sparse — most items appear in a small fraction of transactions. This creates challenges in finding meaningful patterns amidst noise.

Challenge 5: Setting Appropriate Thresholds

- • Too high min_support: Misses rare but important patterns
- • Too low min_support: Generates overwhelming number of rules, most of which are noise
- • Finding the right balance requires domain expertise and iterative experimentation

2. Scalable Methods and Solutions

Solution 1: FP-Growth Algorithm (Tree-Based Compression)

- • Compresses the database into an FP-Tree, requiring only 2 database scans
- • Eliminates candidate generation entirely
- • For our dataset: FP-Tree has at most 5 paths (one per transaction), much smaller than the 7 possible itemsets
- • Scales well to millions of transactions when items are correlated

Solution 2: Partitioning Method

- • Divide the database into non-overlapping partitions
- • Mine each partition independently to find local frequent itemsets
- • Combine results: A globally frequent itemset must be frequent in at least one partition
- • Advantage: Each partition fits in memory; only 2 database scans needed

Solution 3: Sampling

- • Mine a random sample of the database instead of the entire dataset
- • Use a lower support threshold on the sample to compensate for sampling error
- • Significantly faster but may miss some frequent itemsets (false negatives)
- • Can be combined with a verification pass on the full database

Solution 4: Distributed and Parallel Mining

- MapReduce-based Apriori: Distribute candidate counting across multiple nodes
- Spark-based FP-Growth (PFP): Parallel FP-Growth for big data frameworks
- Each node processes a subset of transactions, results are aggregated
- Scales horizontally with more compute resources

Solution 5: Hash-Based Techniques

- Use hash functions to estimate itemset counts
- Park-Chen-Yu (PCY) algorithm: Hash pairs of items to buckets during Pass 1
- Eliminates candidate pairs whose hash bucket count is below the threshold
- Reduces candidate set size in Pass 2

Solution 6: Dynamic Itemset Counting (DIC)

- Starts counting itemsets of different sizes at different points during database scan
- Doesn't wait until full scan to start next level's candidates
- Reduces total number of database passes

3. Summary Comparison Table

Method	DB Scans	Candidates?	Memory	Best For
Apriori	k (per level)	Yes	Moderate	Small-medium datasets
FP-Growth	2	No	Tree in memory	Dense, large datasets
Partitioning	2	Local only	Low per partition	Very large databases
Sampling	1-2	Yes	Low	Quick approximate results
MapReduce	k (distributed)	Yes	Distributed	Big data clusters
PCY/Hashing	2-3	Reduced	Hash tables	Pair counting optimization

SUMMARY OF KEY CONCEPTS

Concept	Definition	Formula/Key Point
Support	Frequency of itemset in database	$\text{Support}(X) = \text{Count}(X) / N$
Confidence	Conditional probability of rule	$\text{Conf}(X \rightarrow Y) = \text{Sup}(X \cup Y) / \text{Sup}(X)$
Lift	Correlation strength measure	$\text{Lift}(X \rightarrow Y) = \text{Conf}(X \rightarrow Y) / \text{Sup}(Y)$
Apriori Property	Infrequent set $\rightarrow$ infrequent superset	Anti-monotone property
FP-Tree	Compressed database representation	Only 2 database scans needed
Constraint-Based	Push constraints into mining	Reduces search space
Multilevel	Mining at different hierarchy levels	Variable support per level
Multidimensional	Rules across multiple attributes	Inter/Hybrid dimensional